Lecture 9. Random Number Generation

Congruential generators, inverse CDF, Box–Muller, acceptance–rejection, multivariate sampling

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9.1 Uniform random numbers

All random number generation starts from i.i.d. Unif$(0,1)$ variates; computers produce deterministic **pseudo-random** sequences that only look random.

Multiplicative congruential generators. For a large prime m and $a \in \{2, \dots, m-1\}$, iterate $x_{k+1} = ax_k \bmod m$ and output $u_k = x_k/m$.

Proposition 9.1

The map $f(x) = ax \bmod m$ is a bijection of $\{1, \dots, m-1\}$. The sequence is periodic with period $n = \min\{k \geq 1 : a^k \equiv 1 \bmod m\}$, which divides $m-1$.

Proof. m is prime, so $\gcd(a, m) = 1$ and $ax \not\equiv 0$ for $1 \leq x \leq m-1$; $ax \equiv ay$ gives $m \mid x - y$, so $x = y$. Thus f is injective, hence bijective. As $x_k = a^k x_0 \bmod m$, the sequence returns to x_0 exactly when $a^k \equiv 1$, and Fermat's little theorem $a^{m-1} \equiv 1$ makes the period divide $m-1$. ■

x_0 is the **seed**. For a **primitive root** a (period $m-1$) the sequence runs through $\{1, \dots, m-1\}$; a classical choice is $m = 2^{31} - 1$, $a = 7^5 = 16807$, period $2^{31} - 2$ (early MATLAB).

Mixed congruential generators. $x_{i+1} = (ax_i + c) \bmod m$, $c \neq 0$, allows $m = 2^M$ (free reduction in binary). By Hull–Dobell the period is the full m iff $\gcd(c, m) = 1$, every prime factor of m divides $a - 1$, and $4 \mid a - 1$ when $4 \mid m$; both $m = 2048$, $a = 1229$, $c = 1$ and the old C-library $m = 2^{32}$, $a = 1103515245$, $c = 12345$ qualify. The first gives only 2048 values (multiples of $1/2048$) and repeats after 2048 draws; the second has poor low-order bits (period 2^k modulo 2^k; the last bit alternates). Plots of (u_i, u_{i+1}) show the lattice structure of both.

A good generator has a long period and passes tests of uniformity and independence. Congruential p-tuples $(u_k, \dots, u_{k+p-1})$ lie on few parallel hyperplanes in $[0,1]^p$ (Marsaglia; RANDU puts all triples on 15 planes); shuffling helps. R's default **Mersenne Twister** (Matsumoto–Nishimura, 1998) has period $2^{19937} - 1$ and is equidistributed in up to 623 dimensions.

9.2 The inverse CDF method

Motivation. If U is uniform, $F^{-1}(U) \sim F$: the simplest sampler when the quantile function is explicit.

For a cdf F define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, $0 < u < 1$.

Theorem 9.2

Let $X \sim F$ and $U \sim \text{Unif}(0,1)$.

(a) $F^{-1}(U)$ has cdf F.

(b) If F is continuous, $F(X) \sim \text{Unif}(0,1)$.

(c) In general, $P\{F(X) \leq y\} \leq y$ for $y \in [0,1]$.

Proof. (a) We show $\{F^{-1}(u) \leq t\} = \{u \leq F(t)\}$. By monotonicity and right continuity, $\{x : F(x) \geq u\} = [F^{-1}(u), \infty)$, so $F(F^{-1}(u)) \geq u$. If $F^{-1}(u) \leq t$ then $u \leq F(F^{-1}(u)) \leq F(t)$; if $u \leq F(t)$ then t is in the set, so $F^{-1}(u) \leq t$. Hence $P(F^{-1}(U) \leq t) = P(U \leq F(t)) = F(t)$.

(b) $P\{F(X) \leq F(t)\} = F(t)$: the event is $\{X \leq t\} \cup \{X > t, F(X) = F(t)\}$, and the second set, where F is flat, has probability 0. For $u \in (0,1)$ continuity gives $F(t) = u$, so $P\{F(X) \leq u\} = u$.

(c) By (a) take $X = F^{-1}(U)$; $F(F^{-1}(U)) \geq U$ gives $\{F(X) \leq y\} \subset \{U \leq y\}$. ■

Example 9.1

- Exponential, $F(x) = 1 - e^{-\lambda x}$: $F^{-1}(u) = -\frac{1}{\lambda}\log(1-u)$, or $-\frac{1}{\lambda}\log U$ as $1 - U \stackrel{d}{=} U$.
- Cauchy(μ, β): $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x-\mu}{\beta}$, $F^{-1}(u) = \mu + \beta\tan(\pi u - \frac{\pi}{2})$.
- Discrete uniform on $\{1, \dots, k\}$: $F^{-1}(u) = \lceil ku \rceil$.
- Geometric(p) on $\{1, 2, \dots\}$: $\lceil Y \rceil$ for $Y \sim \text{Exp}(\lambda)$, $1 - e^{-\lambda} = p$, so $X = \lceil \log U / \log(1-p) \rceil$.

If F^{-1} has no closed form but F (or only f) is computable, solve $F(x) = u$ numerically per draw: bisection, or Newton $x \leftarrow x - \{F(x) - u\}/f(x)$ (Lecture 6), with $F = \int_{-\infty}^{x} f$ by quadrature if needed. This is exact to tolerance but costly, and heavy tails (e.g. t_4) need a wide bracket.

9.3 Normal random numbers

Φ^{-1} has no closed form (accurate rational approximations exist).

Box–Muller. For independent $X, Y \sim N(0,1)$ in polar coordinates (R, Θ) the joint density is $\frac{1}{2\pi}e^{-r^2/2}r$, so $\Theta \sim \text{Unif}(0, 2\pi)$ independently of R, with $P(R^2 > s) = e^{-s/2}$ ($R^2 \sim$ Exp, mean 2). So for independent $U, V \sim \text{Unif}(0,1)$,

$$R = \sqrt{-2\log U}, \quad \Theta = 2\pi V, \qquad (X, Y) = (R\cos\Theta, R\sin\Theta)$$

are independent standard normals.

Marsaglia's polar method. If $(U, V) = (T\cos\Phi, T\sin\Phi)$ is uniform on the unit disk, $P(T^2 \leq t, \Phi \leq \phi) = t\frac{\phi}{2\pi}$, so $T^2 \sim \text{Unif}(0,1)$ and $\Phi \sim \text{Unif}(0, 2\pi)$ independently; $-2\log T^2$ replaces R^2 and $(U/T, V/T)$ replaces $(\cos\Theta, \sin\Theta)$:

(1) draw U, V i.i.d. Unif$(-1, 1)$;
(2) if $T^2 = U^2 + V^2 \geq 1$, return to (1);
(3) output $X = U\sqrt{-2\log T^2/T^2}$ and $Y = V\sqrt{-2\log T^2/T^2}$.

It avoids trigonometric functions at the price of $4/\pi$ times as many pairs (a rejection step, Section 9.5).

9.4 Generating from definitions

- Bernoulli(p): $\mathbf{1}\{U \leq p\}$. Binomial(n, p): sum of n Bernoullis. Negative binomial(r, p): sum of r geometrics.
- Poisson(λ): the N with $\prod_{i=1}^{N} U_i \geq e^{-\lambda} > \prod_{i=1}^{N+1} U_i$, since $T_i = -\log U_i/\lambda$ are Exp(λ) interarrival times and N counts arrivals in $[0,1]$: $T_1 + \cdots + T_N \leq 1 < T_1 + \cdots + T_{N+1}$.
- $\chi^2(\nu)$: $\sum_{i \leq \nu} Z_i^2$; for even ν, also $-2\log\prod_{i \leq \nu/2} U_i$, since $\chi^2(\nu) = \text{Gamma}(\nu/2, 2)$.
- $t(\nu) = Z/\sqrt{X/\nu}$ and $F(\nu_1, \nu_2) = (X_1/\nu_1)/(X_2/\nu_2)$ from independent normal and chi-square variates.

9.5 Acceptance–rejection sampling

Motivation. With a known density up to a constant, sample an easier envelope and accept with probability proportional to the ratio; accepted draws are exact.

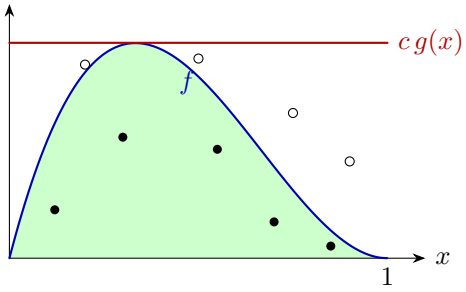


Figure. *Acceptance–rejection for a Beta*$(2,3)$ *target* f *under the uniform envelope* $c\,g$*: filled points are accepted, open ones rejected; the acceptance rate is* $1/c$.

To sample a density f without tractable cdf (von Neumann):

(1) find a density g that is easy to sample and a constant c with $f \le cg =: h$ (an *envelope*);
(2) draw $X \sim g$ and an independent $U \sim \mathrm{Unif}(0,1)$;
(3) accept X if $U \le f(X)/h(X)$; otherwise return to (2).

The **body** of a nonnegative function f is $B_f = \{(x,y) : 0 \le y \le f(x)\}$, with area $|B_f| = \int f$.

Theorem 9.3

If $X \sim g$ and $U \sim \mathrm{Unif}(0,1)$ are independent, then $(X, cg(X)U)$ is uniform on B_{cg}.

Proof. Given $X = x$, $Y = cg(x)U$ has density $\frac{1}{cg(x)}\mathbf{1}\{0 \le y \le cg(x)\}$; times $g(x)$, the joint density is $\frac{1}{c}\mathbf{1}_{B_{cg}}$, and $|B_{cg}| = c$. ■

Corollary 9.4

An accepted X has density f, and each proposal is accepted with probability $1/c$.

Proof. Acceptance is $(X,Y) \in B_f \subset B_{cg}$, so the accepted (X,Y) is uniform on B_f, with x-marginal $\int_0^{f(x)} dy/|B_f| = f(x)$. The acceptance probability is $|B_f|/|B_{cg}| = 1/c$. ■

So the envelope should be tight ($c \approx 1$); the number of proposals per draw is geometric with mean c. f need only be known up to a constant shared with the envelope.

Example 9.2 (Uniform on the disk)

In Marsaglia's method $f = \frac{1}{\pi}\mathbf{1}\{u^2+v^2<1\}$, $g = \frac{1}{4}\mathbf{1}_{[-1,1]^2}$, $c = 4/\pi$, and $f/(cg)$ is the disk indicator: accept iff $U^2 + V^2 < 1$.

Example 9.3 (Beta variates from uniforms)

For $\mathrm{Beta}(a,b)$, $a,b>1$, the density is bounded by its mode value $d_{\max}$ at $(a-1)/(a+b-2)$. With $g = \mathrm{Unif}(0,1)$, $c = d_{\max}$: keep $Y \sim \mathrm{Unif}(0,1)$ iff $U \le f(Y)/d_{\max}$, at acceptance

rate $1/d_{\max}$ (1000 proposals give about $1000/d_{\max}$ draws). Simple, but wasteful for peaked densities (large a, b).

Example 9.4 (Gamma variates)

As $c\,\text{Gamma}(\alpha,1) = \text{Gamma}(\alpha,c)$ and $\text{Gamma}(\alpha,1) \stackrel{d}{=} \text{Gamma}(\lfloor\alpha\rfloor,1) \oplus \text{Gamma}(\alpha-\lfloor\alpha\rfloor,1)$ (the first a sum of exponentials), take $0<\alpha<1$. Then $x^{\alpha-1}e^{-x} \le x^{\alpha-1}$ on $[0,1]$ and $\le e^{-x}$ on $(1,\infty)$, giving $h(x) = \{x^{\alpha-1}\mathbf{1}_{[0,1]} + e^{-x}\mathbf{1}_{(1,\infty)}\}/\Gamma(\alpha)$, $c = (\frac{1}{\alpha}+\frac{1}{e})/\Gamma(\alpha)$. $g = h/c$ has cdf $G(x) = x^\alpha/(1+\alpha/e)$ on $[0,1]$ and $1-\alpha e^{-x}/(1+\alpha/e)$ for $x>1$, with closed-form inverse

$$G^{-1}(u) = \begin{cases} \{(1+\alpha/e)u\}^{1/\alpha}, & u \le 1/(1+\alpha/e), \\ -\log\{(1/\alpha+1/e)(1-u)\}, & \text{otherwise.} \end{cases}$$

Example 9.5 (Gamma with $\alpha > 1$: a Cauchy envelope)

Let $\tilde f(x) = x^{\alpha-1}e^{-x}$ $(x>0)$, $\alpha>1$, $\mu = \alpha-1$, $\sigma^2 = 2\alpha-1$, and

$$g(x) = \frac{\mu^\mu e^{-\mu}}{1+(x-\mu)^2/\sigma^2},$$

a multiple of the Cauchy(μ,σ) density. *Domination*: $k(x) = x^{\alpha-1}e^{-x}\{\sigma^2+(x-\mu)^2\}$ has $k'(x) = -x^{\alpha-2}e^{-x}(x-\alpha)^2(x-\mu)$, ≥ 0 for $x<\mu$ and ≤ 0 for $x>\mu$, so $k \le k(\mu) = \sigma^2\mu^\mu e^{-\mu}$, i.e. $\tilde f \le g$ on $(0,\infty)$. *Algorithm*: $Y = \mu + \sigma\tan\{\pi(V-\frac{1}{2})\}$, $V \sim \text{Unif}(0,1)$; reject if $Y \le 0$; else accept if $U \le \tilde f(Y)/g(Y)$. *Efficiency*: $\int g = \pi\sigma\mu^\mu e^{-\mu}$ and $\int \tilde f = \Gamma(\alpha)$, so the acceptance probability is

$$\frac{\Gamma(\alpha)}{\pi\sqrt{2\alpha-1}\,(\alpha-1)^{\alpha-1}e^{-(\alpha-1)}} \approx 0.47,\ 0.50,\ 0.54,\ 0.56 \quad (\alpha = 1.5, 2, 5, 100),$$

increasing to $1/\sqrt{\pi} \approx 0.564$ by Stirling, $\Gamma(\mu+1) \sim \sqrt{2\pi\mu}\,\mu^\mu e^{-\mu}$: about two proposals per draw for every α.

9.6 Multivariate distributions

Batch methods.

- Multinomial$(n; p_1,\dots,p_k)$: draw n categories (j when $U \in [\sum_{l<j} p_l, \sum_{l\le j} p_l)$) and count. (Independent Poissons λp_j conditioned on sum n also work, inefficiently.)
- $N(\mu,\Sigma)$: $\mu + LZ$, $Z \sim N(0, I_p)$, $LL^\top = \Sigma$: Cholesky (cheapest) or $Q\Lambda^{1/2}Q^\top$ (handles singular Σ). For large p the factorization is the bottleneck.
- Multivariate $t_\nu(\mu,\Sigma)$: $\mu + X/\sqrt{W/\nu}$ with $X \sim N(0,\Sigma)$ independent of $W \sim \chi^2(\nu)$.

Sequential (conditional) sampling. Draw components one at a time from their conditionals:

- Multinomial: $X_1 \sim B(n, p_1)$, then $X_2 \mid X_1 = x_1 \sim B(n-x_1, p_2/(1-p_1))$, and so on.
- Multivariate normal: $X_1 \sim N(\mu_1, \sigma_{11})$, then $(X_2,\dots,X_p) \mid X_1 = x_1 \sim N(\mu_2 + \Sigma_{21}(x_1-\mu_1)/\sigma_{11},\ \Sigma_{22} - \Sigma_{21}\Sigma_{12}/\sigma_{11})$, and recurse.

9.7 Selected exercises

Motivation. Weak low-order bits of power-of-two congruential generators, and an optimal normal sampler by rejection.

Exercise 9.1 (Low-order bits of a mixed congruential generator)

Let $x_{i+1} = (ax_i + c) \bmod 2^M$ and, for $1 \le k \le M$, $y_i = x_i \bmod 2^k$ (the k lowest bits of x_i).
(a) Show that $y_{i+1} = (ay_i + c) \bmod 2^k$, and conclude that (y_i) is eventually periodic with period at most 2^k.
(b) Show that if a and c are odd, the last bit of x_i alternates between 0 and 1.

Solution. [Supplement]

(a) $x_{i+1} = ax_i + c - q2^M$ and $2^k \mid 2^M$, so $x_{i+1} \equiv ax_i + c \equiv ay_i + c \pmod{2^k}$: $y_{i+1} = (ay_i + c) \bmod 2^k$, a fixed function of y_i with at most 2^k values. Two of $y_0, \dots, y_{2^k}$ coincide, $y_s = y_{s+d}$ with $1 \le d \le 2^k$, and then $y_{i+d} = y_i$ for $i \ge s$.

(b) $k = 1$: a, c odd give $y_{i+1} = (y_i + 1) \bmod 2$.

So for $m = 2^{32}$, $a = 1103515245$, $c = 12345$ the last k bits repeat with period at most 2^k despite full period 2^{32}: use the high-order bits ($u_i = x_i/2^M$), never $x_i \bmod 2$ as a coin. ■

Exercise 9.2 (Normal variates from a Laplace envelope)

Let ϕ be the $N(0,1)$ density and $g_\lambda(x) = \frac{\lambda}{2} e^{-\lambda|x|}$ the Laplace density, $\lambda > 0$.
(a) Find the smallest $c(\lambda)$ with $\phi \le c(\lambda) g_\lambda$.
(b) Find the λ that maximizes the acceptance probability, and that probability.
(c) Show that for the optimal λ the sampler can be run with exponential variates only.

Solution. [Supplement]

(a) $\phi(x)/g_\lambda(x) = \frac{2}{\lambda\sqrt{2\pi}} \exp(-\frac{x^2}{2} + \lambda|x|)$, and $-\frac{t^2}{2} + \lambda t$ is maximized over $t \ge 0$ at $t = \lambda$, value $\lambda^2/2$:

$$c(\lambda) = \sup_x \frac{\phi(x)}{g_\lambda(x)} = \sqrt{\frac{2}{\pi}} \frac{e^{\lambda^2/2}}{\lambda}.$$

(b) Maximize $1/c(\lambda)$ (Corollary 9.4): $\log c(\lambda) = \text{const} + \frac{\lambda^2}{2} - \log\lambda$ has derivative $\lambda - 1/\lambda$ and second derivative $1 + \lambda^{-2} > 0$, so $\lambda = 1$, $c(1) = \sqrt{2e/\pi} \approx 1.3155$, acceptance ≈ 0.760.

(c) $\phi(x)/\{c(1)g_1(x)\} = \exp\{-\frac{1}{2}(|x| - 1)^2\}$. With $X = SE_1$ ($E_1 \sim \text{Exp}(1)$, independent random sign S), $U \le e^{-(E_1-1)^2/2}$ iff $E_2 \ge (E_1 - 1)^2/2$ for $E_2 = -\log U \sim \text{Exp}(1)$. So draw E_1, E_2 until $E_2 \ge (E_1 - 1)^2/2$ and return $\pm E_1$; on average 1.3155 pairs per variate. ■